
MA 549: Abstract Algebra

Course Code	MA 549
Course Name	Abstract Algebra
L-T-P-C	3-0-0-3
Prerequisites	NA
Intended for	UG 3rd and 4th year students/PG
Distribution	Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Module I** [21 Lectures]

Binary operation, and its properties, Definition of a group, Examples and basic properties. Subgroups, Coset of a subgroup, Lagrange's theorem. Cyclic groups, Order of a group. Normal subgroups, Quotient group. Homomorphisms, Kernel Image of a homomorphism, Isomorphism theorems. Permutation groups, Cayley's theorems. Direct product of groups. Structure of finite abelian groups. Applications, Private and public key cryptography, some nontrivial examples

- **Module II** [7 Lectures]

Rings: definition, Examples and basic properties. Zero divisors, Integral domains, Fields, Characteristic of a ring, Quotient field of an integral domain. Subrings, Ideals, Quotient rings, Isomorphism theorems. Ring of polynomials. Prime, Irreducible elements and their properties, UFD, PID and Euclidean domains. Prime ideal, Maximal ideals. Extension fields, Algebraic extensions, Finite fields.

Text Books

1. Joseph Gallian, **Contemporary Abstract Algebra**, 7th Edition, Brooks Cole, 2009.
2. J. B. Fraleigh, **A first Course in Abstract Algebra**, Narosa Publishing House, 2003.

References

1. I. N. Herstein, **Topics in Algebra**, Wiley Eastern Ltd., 1975.
2. Klima, Sigmon and Stitzinger, **Applications of Abstract Algebra with Maple and Matlab**, 2nd Edition, 2006.